

iStar-DT-X ver 1.0

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iStarDT-X Grammar (1/3)

Primitives, Enums

```
[symbol]      ::= Any valid XML character except quotes and angle brackets
[letter]      ::= "A" | "B" | ... | "Z" | "a" | "b" | ... | "z"
[letter-small] ::= "a" | "b" | ... | "z"
[digit]       ::= "0" | "1" | ... | "9"
[character]   ::= [letter] | [digit] | [symbol]
[alphanum]    ::= [letter] | [digit] | "."
[string]      ::= [character]+
[identifier]  ::= [letter-small][alphanum]*
[boolean]     ::= "true" | "false"
[decimal]     ::= [digit]+ [ "." [digit]+ ]
[integer]     ::= [digit]+
```

Boolean Expressions

```
[boolean-expression] ::= [bool-const] | [atom] | [numeric-comparison] | [boolean-operation]
[bool-const]         ::= "<boolConst>" [boolean] "</boolConst>"
[atom]               ::= [predicate-ref] | [goal-ref] | [task-ref]
[boolean-operation]  ::= [and] | [or] | [not] | [prev]
[and]                ::= "<and>" [boolean-expression] [boolean-expression]+ "</and>"
[or]                 ::= "<or>" [boolean-expression] [boolean-expression]+ "</or>"
[not]                ::= "<not>" [boolean-expression] "</not>"
[implies]            ::= "<implies>" [boolean-expression] [boolean-expression] "</implies>"
[prev]               ::= "<previous>" [atom] "</previous>"
[comparison-op]      ::= "gt" | "gte" | "lt" | "lte" | "eq" | "neq"
[numeric-comparison] ::= "<" [comparison-op] ">"
                    " <left>" [numeric-expression] "</left>"
                    " <right>" [numeric-expression] "</right>"
                    "</" [comparison-op] ">"
```

Numeric Expressions

```
[numeric-expression] ::= [num-const] | [num-atom] | [num-previous] |
                        [arithmetic-operation]
[num-const]          ::= "<numConst>" [decimal] "</numConst>"
[num-atom]           ::= [atom] | [variable-id] | [qual-id]
[variable-id]        ::= "<variableID>" [identifier] "</variableID>"
[qual-id]            ::= "<qualID>" [identifier] "</qualID>"
[num-previous]       ::= "<previous>" [num-atom] "</previous>"
[arithmetic-operator] ::= [add] | [subtract] | [multiply] | [divide]
[add]                ::= "<add>" [numeric-expression] [numeric-expression]+ "</add>"
[multiply]           ::= "<multiply>" [numeric-expression] [numeric-expression]+
                    "</multiply>"
[subtract]           ::= "<subtract>" " <left>" [numeric-expression] "</left>"
                    " <right>" [numeric-expression] "</right>" "</subtract>"
[divide]             ::= "<divide>" " <left>" [numeric-expression] "</left>"
                    " <right>" [numeric-expression] "</right>" "</divide>"
```

Parameters

```
[parameters] ::= "<parameters>" [parameter]+ "</parameters>"
[parameter]  ::= "<parameter>" [param-type] [param-name] "</parameter>"
[param-type] ::= "<param-type>" [identifier] "</param-type>"
[param-name] ::= "<param-name>" [identifier] "</param-name>"
```

Temporal Folumae

```
[temporal-formula] ::= [boolean-expression]
[temporal-formula] ::= "<always>" [temp-form] "</always>"
[temporal-formula] ::= "<eventually>" [temp-form] "</eventually>"
[temporal-formula] ::= "<next>" [temp-form] "</next>"
[temporal-formula] ::= "<until>" [temp-form] [temp-form] "</until>"
```

Figure 1: BNF specification of iStarDT-X: primitives and expressions.

iStarDT-X Grammar (2/3)

Root

```

[istardt-model] ::= "<iStarDT " [namespaces] ">" [model-header] [options] [actors]
                  "</iStarDT>"
[namespaces]    ::= " xmlns=" [string] "' " [" xmlns:xsi=" [string] "' ]
                  [ " xsi:schemaLocation=" [string] "' " ]
[model-header]  ::= "<header title = " [string] "' author = " [string]
                  "' source = " [string] "' lastUpdated=" [string] "'>"
                  [string] "</header>"
[options]       ::= "<options continuous = " [boolean] "'
                  infeasibleActionPenalty = " [decimal] "'>
                  </options>"
[actors]        ::= "<actors>" [actor]+ "</actors>"
[actor]         ::= "<actor " [actor-attributes] ">" [actor-content] "</actor>"
[actor-attributes] ::= "name=" [identifier] "' " [description=" [string] "' ]
[actor-content]  ::= [ [goals] ] [ [tasks] ] [ [qualities] ] [ [predicates] ]
                  [ [variables] ] [ [condBoxes] ] [ [cross-runs] ]
                  [ [exported-set] ] [ [initializations] ] [ [definitions] ]
                  [ [invars] ]

```

Goals

```

[goals]         ::= "<goals>" [goal]+ "</goals>"
[goal]          ::= "<goal name=" [identifier] "' mode = " [goal-mode]
                  [" root = " [boolean] "' ] [" activation = " [act-style] " ]
                  [ " description=" [string] "' ] [ " terminal=" [boolean] "' ]
                  [ " episodeLength=" [integer] "' ] [" actor=" [identifier] "' ]
                  [goal-content] "</goal>"
[goal-mode]     ::= 'achieve' | 'maintain' | 'avoid'
[act-style]     ::= 'called' | 'triggered'
[goal-content]  ::= [ [parameters] ] [ [pre] ] [ [tri] ] [ [goal-defn] ] [ [refinement] ]
[pre]           ::= "<pre>" [boolean-expression] "</pre>"
[tri]           ::= "<tri>" [boolean-expression] "</tri>"
[goal-defn]     ::= "<goal-defn>" [activation] [fulfillment] "</goal-defn>"
[activation]    ::= "<activation>" [boolean-expression] "</activation>"
[fulfillment]  ::= "<fulfillment>" [boolean-expression] "</fulfillment>"
[refinement]    ::= "<refinement type=" [ref-type] "'>"
                  [ [ref-condition] ]
                  ([childGoal] | [childTask])+ "</refinement>"
[ref-type]      ::= "AND" | "OR"
[ref-condition] ::= "<ref-condition>" [pick-cond] | [forall-cond] "</ref-condition>"
[pick-cond]     ::= "<pick>" [cond-content] "</pick>"
[forall-cond]   ::= "<forall>" [cond-content] "</forall>"
[cond-content]  ::= "<params>" [cond-param]+ "</params>"
                  "<condition>" [predicate-ref] "</condition>"
[cond-param]    ::= "<param>" [identifier] "</param>"
[childGoal]     ::= [goal-ref]
[childTask]     ::= [task-ref]
[goal-ref]      ::= "<goal name=" [identifier] "'>" [parameters] "</goal>"

```

Tasks

```

[tasks]         ::= "<tasks>" [task]+ "</tasks>"
[task]          ::= "<task name=" [identifier] "' " [ " description=" [string] "' ]
                  [ " actor=" [identifier] "' ] [" delegation-style=" [del-style] " ]
                  ">" [task-content] "</task>"
[task-content]  ::= [ [parameters] ] [effectGroup] [ [pre] ] [ [tri] ]
[effectGroup]   ::= "<effectGroup>" [effect]+ "</effectGroup>"
[effect]        ::= "<effect name=" [identifier] "' " [ " satisfying=" [boolean] "' ]
                  [ " probability=" [decimal] "' ] [ " description=" [string] "' ]
                  ">" [effect-content] "</effect>"
[effect-content] ::= [ [turnsTrue]* ] [ [turnsFalse]* ] [ [sets]* ] [ [pre] ] [ [npr] ]
[turnsTrue]     ::= "<turnsTrue>" [predicate-ref] "</turnsTrue>"
[turnsFalse]    ::= "<turnsFalse>" [predicate-ref] "</turnsFalse>"
[sets]          ::= "<set>" [variable-id] [num-const] "</set>"
[task-ref]      ::= "<task name=" [identifier] "'>" [parameters] "</task>"

```

Figure 2: BNF specification of iStarDT-X: Models, Goals, and Tasks.

iStarDT-X Grammar (3/3)

Qualities

```
[qualities] ::= "<qualities>" [quality]+ "</qualities>"
[quality]   ::= "<quality name='\" [identifier] '\" description='\" [string] '\"\"
                [\" root='\" [boolean] '\"\" \">\"
                [numeric-expression] "</quality>"
```

Predicates and Variables

```
[predicates] ::= "<predicates>" [predicate]+ "</predicates>"
[predicate]  ::= "<predicate name = '\" [identifier] '\" description = '\" [string] '\">\"
                [ [parameters] ] "</predicate>" % see above
[predicate-ref] ::= "<predicate name = '\" [identifier] '\">\"
                [ [parameters] ] "</predicate>" % see above
[variables]  ::= "<variables>" [variable]+ "</variables>"
[variable]   ::= "<variable description = '\" [string] '\">\"
                [identifier] "</variable>"
```

Condition Boxes

```
[condBoxes] ::= "<condBoxes>" [condBox]+ "</condBoxes>"
[condBox]   ::= "<condBox name='\" [identifier] '\" description='\" <string> '\"\" \">\"
                [predicate-ref] "</condBox>"
```

Cross-runs, Exported Sets, Initializations, Defininitios

```
[cross-runs] ::= "<crossRuns>" [cross-run]+ "</crossRuns>"
[cross-run]  ::= "<crossRun>" [predicate-id] | [variable-id] | [qual-id]
                "</crossRun>"
[exported-set] ::= "<exportedSet>" [export]+ "</exportedSet>"
[export]     ::= "<export continuous='\" [boolean] [\" minVal='\" [decimal]
                [\" maxVal='\" [decimal] ] \">\" [exp-id] "</export>"
[exp-id]     ::= [goal-id] | [task-id] | [predicate-id] | [variable-id] | [qual-id]
[initializations] ::= "<initializations>" [initialization] "</initializations>"
[initialization] ::= [b-init] | [d-init]
[b-init]     ::= "<initialization element='\" [predicate-id] \"> \"
                [boolean] "</initialization>"
[d-init]     ::= "<initialization element='\" [cont-init-id] \"> \"
                [decimal] "</initialization>"
[cont-init-id] ::= [variable-id] | [qual-id]
[definitions] ::= "<definitions>" [definition] "</definitions>"
[definition]  ::= "<definition>" [definiendum] [definiens] "</definition>"
[definiendum] ::= "<definiendum name = '\" [identifier] '\">\"
                [ [parameters] ] "</definiendum>"
[definiens]   ::= "<definiens>" [boolean-expression] "</definiens>"
```

Invariants

```
[invariants] ::= "<invariants>" [invariant] "</invariants>"
[invariant]   ::= "<invariant>" [temporal-formula] "</invariant>"
```

Figure 3: BNF specification of iStarDT-X: Qualities, Predicates, Variables, Condition Boxes, Cross-run, Exported and Initialization sets.